

lab11(SSU)

October 13, 2025

1 Lab 11: Simulations

We will go over [iteration](#) and [simulations](#), as well as introduce the concept of [randomness](#).

The data used in this lab will contain salary data and other statistics for basketball players from the 2014-2015 NBA season. This data was collected from the following sports analytic sites: [Basketball Reference](#) and [Spotrac](#).

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install py pandoc
      !pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntaxtex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex

texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 54 newly installed, 0 to remove and 38 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17
[33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]

Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]

Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]

Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all

20180621-3.1 [10.2 MB]
 Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
 all 18-1 [4,720 B]
 Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
 java all 43-1 [10.8 kB]
 Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
 java all 1.2-2 [60.3 kB]
 Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
 all 1.18 [5,336 B]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
 all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
 [5,100 B]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]
 Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
 all 12.2-1ubuntu1 [185 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64
 1.41-4build2 [61.3 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64

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2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 2s (95.0 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126675 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...

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Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...

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Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...

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Selecting previously unselected package xfonts-utils.
 Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
 Unpacking xfonts-utils (1:7.7+6build2) ...
 Selecting previously unselected package lmodern.
 Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
 Unpacking lmodern (2.004.5-6.1) ...
 Selecting previously unselected package preview-latex-style.
 Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
 Unpacking preview-latex-style (12.2-1ubuntu1) ...
 Selecting previously unselected package t1utils.
 Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
 Unpacking t1utils (1.41-4build2) ...
 Selecting previously unselected package teckit.
 Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
 Unpacking teckit (2.5.11+ds1-1) ...
 Selecting previously unselected package tex-gyre.
 Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
 Unpacking tex-gyre (20180621-3.1) ...
 Selecting previously unselected package texlive-binaries.
 Preparing to unpack .../41-texlive-binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
 Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
 Selecting previously unselected package texlive-base.
 Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
 Unpacking texlive-base (2021.20220204-1) ...
 Selecting previously unselected package texlive-fonts-recommended.
 Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-fonts-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-base.
 Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-base (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-recommended.
 Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
 Unpacking texlive-latex-recommended (2021.20220204-1) ...
 Selecting previously unselected package texlive.
 Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
 Unpacking texlive (2021.20220204-1) ...
 Selecting previously unselected package libfontbox-java.
 Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libfontbox-java (1:1.8.16-2) ...
 Selecting previously unselected package libpdfbox-java.
 Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
 Unpacking libpdfbox-java (1:1.8.16-2) ...
 Selecting previously unselected package texlive-pictures.
 Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
 Unpacking texlive-pictures (2021.20220204-1) ...
 Selecting previously unselected package texlive-latex-extra.
 Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...

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Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...

```



```

update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST..
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN..
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

```

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link
/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running upmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

 This may take some time... done.

Collecting py pandoc

 Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)

 Downloading py pandoc-1.15-py3-none-any.whl (21 kB)

Installing collected packages: py pandoc

Successfully installed py pandoc-1.15

Collecting datascience

 Downloading datascience-0.18.0-py3-none-any.whl.metadata (910 bytes)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from datascience) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from datascience) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.2)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from datascience) (0.8.2)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Collecting jedi>=0.16 (from ipython->datascience)

 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)

Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)

Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)

Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (2.19.2)

Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.0)

```

Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.10.5)
Downloading datascience-0.18.0-py3-none-any.whl (725 kB)
725.0/725.0 kB
12.4 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
56.2 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.0 jedi-0.19.2

```

```

[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

```

```
import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Run this cell, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
```

1.1 1. Nachos and Conditionals

In Python, the boolean is a data type with only two possible values: **True** and **False**. Expressions containing comparison operators such as **<** (less than), **>** (greater than), and **==** (equal to) evaluate to Boolean values. A list of common comparison operators can be found below!

Run the cell below to see an example of a comparison operator in action.

```
[4]: 3 > (1 + 1)
```

[4]: True

We can even assign the result of a comparison operation to a variable. Note that **==** and **=** are **not** the same!

```
[5]: result = 10 / 2 == 5
result
```

[5]: True

Just like arithmetic operators can be applied on every item of an array, comparison operators can also be used on arrays to compare an entire array with some value. The output of this comparison is an array of boolean values.

```
[6]: make_array(1, 5, 7, 8, 3, -1) > 3
```

[6]: array([False, True, True, True, False, False], dtype=bool)

One day, when you come home after a long week, you see a hot bowl of nachos waiting on the dining table! Let's say that whenever you take a nacho from the bowl, it will either have only **cheese**, only **salsa**, **both** cheese and salsa, or **neither** cheese nor salsa (a sad tortilla chip indeed).

Let's try and simulate taking nachos from the bowl at random using the function, `np.random.choice(...)`.

1.1.1 `np.random.choice`

`np.random.choice` picks one item at random from the given array. It is equally likely to pick any of the items. Run the cell below several times, and observe how the results change. *Tip:* To keep running a cell multiple times you can use the keyboard shortcut `ctrl + return`.

```
[7]: nachos = make_array('cheese', 'salsa', 'both', 'neither')
     np.random.choice(nachos)
```

```
[7]: 'salsa'
```

To repeat this process multiple times, pass in an int `n` as the second argument to return `n` different random choices. By default, `np.random.choice` samples **with replacement** and returns an *array* of items. Sampling **with replacement** means that after an element is drawn, it is replaced back to where you are sampling from and can be drawn again in the future. If we sample `n` times with replacement, each time, every element has an equal chance of being selected.

Run the next cell to see an example of sampling with replacement 10 times from the `nachos` array.

```
[8]: np.random.choice(nachos, 10)
```

```
[8]: array(['salsa', 'salsa', 'salsa', 'neither', 'neither', 'neither',
          'neither', 'salsa', 'both', 'salsa'],
          dtype='<U7')
```

To count the number of times a certain type of nacho is randomly chosen, we can use `np.count_nonzero`

1.1.2 `np.count_nonzero`

`np.count_nonzero` counts the number of non-zero values that appear in an array. When an array of boolean values are passed through the function, it will count the number of `True` values (remember that in Python, **True is coded as 1 and False is coded as 0.**)

Run the next cell to see an example that uses `np.count_nonzero`.

```
[9]: np.count_nonzero(make_array(True, False, False, True, True))
```

```
[9]: 3
```

Question 1.1 Assume we took ten nachos at random, and stored the results in an array called `ten_nachos` as done below. **Find the number of nachos with only cheese using code** (do not manually enter the final answer).

Hint: Our solution involves a comparison operator (e.g. `==`, `<`, ...) and the `np.count_nonzero` method.

```
[11]: ten_nachos = make_array('neither', 'cheese', 'both', 'both', 'cheese', 'salsa',  
    ↪ 'both', 'neither', 'cheese', 'both')  
number_cheese = np.count_nonzero(ten_nachos == 'cheese')  
number_cheese
```

```
[11]: 3
```

Conditional Statements

A conditional statement is a multi-line statement that allows Python to choose among different alternatives based on the truth value of an expression.

Here is a basic example.

```
def sign(x):  
    if x > 0:  
        return 'Positive'  
    else:  
        return 'Negative'
```

If the input `x` is greater than 0, we return the string `'Positive'`. Otherwise, we return `'Negative'`.

If we want to test multiple conditions at once, we use the following general format.

```
if <if expression>:  
    <if body>  
elif <elif expression 0>:  
    <elif body 0>  
elif <elif expression 1>:  
    <elif body 1>  
...  
else:  
    <else body>
```

Only the body for the first conditional expression that is true will be evaluated. Each `if` and `elif` expression is evaluated and considered in order, starting at the top. `elif` can only be used if an `if` clause precedes it. As soon as a true value is found, the corresponding body is executed, and the rest of the conditional statement is skipped. If none of the `if` or `elif` expressions are true, then the `else` body is executed.

For more examples and explanation, refer to the section on conditional statements [here](#).

Question 1.2 Complete the following conditional statement so that the string `'More please'` is assigned to the variable `say_please` if the number of nachos with cheese in `ten_nachos` is less than 5. Use the `if` statement to do this (do not directly reassign the variable `say_please`).

Hint: You should be using `number_cheese` from Question 1.

```
[13]: say_please = '?'  
  
if number_cheese < 5:  
    say_please = 'More please'
```

```
say_please
```

```
[13]: 'More please'
```

Question 1.3 Write a function called `nacho_reaction` that returns a reaction (as a string) based on the type of nacho passed in as an argument. Use the table below to match the nacho type to the appropriate reaction.

Nacho Type	Reaction
cheese	Cheesy!
salsa	Spicy!
both	Wow!
neither	Meh.

Hint: If you're failing the test, double check the spelling of your reactions.

```
[15]: def nacho_reaction(nacho):  
    if nacho == "cheese":  
        return 'Cheesy!'  
    elif nacho == "salsa":  
        return 'Spicy!'  
    elif nacho == "both":  
        return 'Wow!'  
    else:  
        return 'Meh.'  
  
spicy_nacho = nacho_reaction('salsa')  
spicy_nacho
```

```
[15]: 'Spicy!'
```

Question 1.4 Create a table `ten_nachos_reactions` that consists of the nachos in `ten_nachos` as well as the reactions for each of those nachos. The columns should be called `Nachos` and `Reactions`.

Hint: Consider using the `apply` method, which returns an array.

```
[16]: ten_nachos_tbl = Table().with_column('Nachos', ten_nachos)  
ten_nachos_reactions = ten_nachos_tbl.with_column('Reactions', ten_nachos_tbl.  
    ↪ apply(nacho_reaction, 'Nachos'))  
ten_nachos_reactions
```

```
[16]: Nachos | Reactions  
neither | Meh.  
cheese  | Cheesy!  
both    | Wow!  
both    | Wow!  
cheese  | Cheesy!
```



```
salsa    | Spicy!
both     | Wow!
neither  | Meh.
cheese   | Cheesy!
both     | Wow!
```

Question 1.5 Using code, find the number of ‘Wow!’ reactions for the nachos in `ten_nachos_reactions`.

```
[18]: number_wow_reactions = np.count_nonzero(ten_nachos_reactions.
      ↪column('Reactions') == 'Wow!')
      number_wow_reactions
```

```
[18]: 4
```

1.2 2. Simulations and For Loops

Using a `for` statement, we can perform a task multiple times. This is known as iteration. The general structure of a `for` loop is:

`for <placeholder> in <array>:` followed by indented lines of code that are repeated for each element of the `array` being iterated over. You can read more about `for` loops [here](#).

NOTE: We often use `i` as the `placeholder` in our class examples, but you could name it anything! Some examples can be found below.

One use of iteration is to loop through a set of values. For instance, we can print out all of the colors of the rainbow.

```
[19]: rainbow = make_array("red", "orange", "yellow", "green", "blue", "indigo",
      ↪"violet")

      for color in rainbow:
          print(color)
```

```
red
orange
yellow
green
blue
indigo
violet
```

We can see that the indented part of the `for` loop, known as the body, is executed once for each item in `rainbow`. The name `color` is assigned to the next value in `rainbow` at the start of each iteration. Note that the name `color` is arbitrary; we could easily have named it something else. Whichever name we pick, we need to use it consistently throughout the `for` loop.

```
[20]: for another_name in rainbow:
      print(another_name)
```

```
red
orange
yellow
green
blue
indigo
violet
```

In general, however, we would like the variable name to be somewhat informative.

Question 2.1 In the following cell, we’ve loaded the text of *Pride and Prejudice* by Jane Austen, split it into individual words, and stored these words in an array `p_and_p_words`. Using a `for` loop, assign `longer_than_five` to the number of words in the novel that are more than 5 letters long.

Hint: You can find the number of letters in a word with the `len` function.

Hint: How can you use `longer_than_five` to keep track of the number of words that are more than five letters long?

```
[22]: austen_string = open('./DS/Austen_PrideAndPrejudice.txt', encoding='utf-8').
      ↪read()
      p_and_p_words = np.array(austen_string.split())

      longer_than_five = 0

      for a in p_and_p_words:
          if len(a) > 5:
              longer_than_five += 1
      longer_than_five
```

```
[22]: 35453
```

Another way we can use `for` loops is to repeat lines of code many times. Recall the structure of a `for` loop:

`for <placeholder> in <array>:` followed by indented lines of code that are repeated for each element of the array being iterated over.

Sometimes, we don’t care about what the value of the placeholder is. We instead take advantage of the fact that the `for` loop will repeat as many times as the length of our array. In the following cell, we iterate through an array of length 5 and print out “Hello, world!” in each iteration, but we don’t need to use the placeholder `i` in the body of our `for` loop.

```
[24]: for i in np.arange(5):
      ↪print("Hello, world!")
```

```
Hello, world!
Hello, world!
Hello, world!
```

Hello, world!
Hello, world!

Question 2.2 Using a simulation with 10,000 trials, assign `num_different` to the **number** of times, in 10,000 trials, that two words picked uniformly at random (with replacement) from *Pride and Prejudice* have different lengths.

Hint 1: What function did we use in section 1 to sample at random with replacement from an array?

Hint 2: Remember that `!=` checks for non-equality between two items.

```
[31]: trials = 10000
      num_different = 0

      for _ in np.arange(trials):
          word1 = np.random.choice(p_and_p_words)
          word2 = np.random.choice(p_and_p_words)
          if len(word1) != len(word2):
              num_different += 1
      num_different
```

[31]: 8568

1.3 3. Sampling Basketball Data

We will now introduce the topic of sampling, which we'll be discussing in more depth in this week's lectures. We'll guide you through this code, but if you wish to read more about different kinds of samples before attempting this question, you can check out [section 10 of the textbook](#).

Run the cell below to load player and salary data that we will use for our sampling.

```
[32]: player_data = Table().read_table("./DS/player_data.csv")
      salary_data = Table().read_table("./DS/salary_data.csv")
      full_data = salary_data.join("PlayerName", player_data, "Name")

      # The show method immediately displays the contents of a table.
      # This way, we can display the top of two tables using a single cell.
      player_data.show(3)
      salary_data.show(3)
      full_data.show(3)
```

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

<IPython.core.display.HTML object>

Rather than getting data on every player (as in the tables loaded above), imagine that we had gotten data on only a smaller subset of the players. For 492 players, it's not so unreasonable to expect to see all the data, but usually we aren't so lucky.

If we want to make estimates about a certain numerical property of the population, we may have to come up with these estimates based only on a smaller sample. The numerical property of the population is known as a **parameter**, and the estimate is known as a **statistic** (e.g. the mean or median). Whether these estimates are useful or not often depends on how the sample was gathered. We have prepared some example sample datasets to see how they compare to the full NBA dataset. Later we'll ask you to create your own samples to see how they behave.

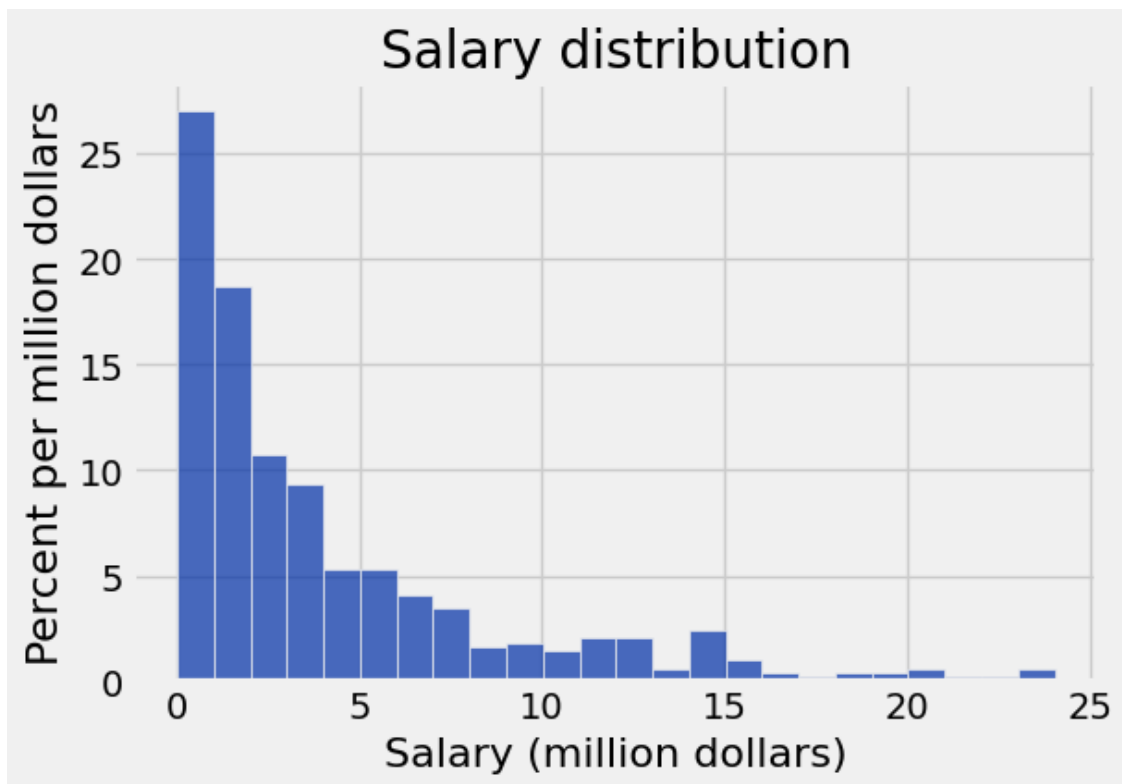
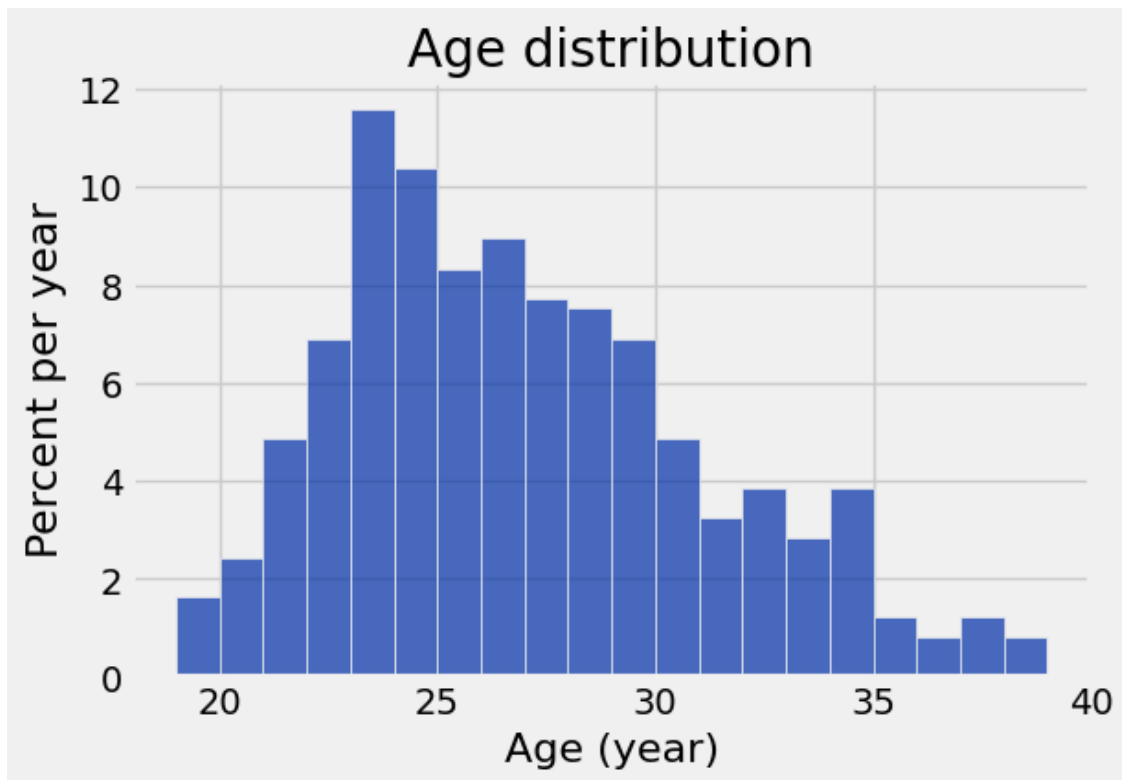
To save typing and increase the clarity of your code, we will package the analysis code into a few functions. This will be useful in the rest of the lab as we will repeatedly need to create histograms and collect summary statistics from that data.

We've defined the `histograms` function below, which takes a table with columns `Age` and `Salary` and draws a histogram for each one. It uses bin widths of 1 year for `Age` and \$1,000,000 for `Salary`.

```
[33]: def histograms(t):
    ages = t.column('Age')
    salaries = t.column('Salary')/1000000
    t1 = t.drop('Salary').with_column('Salary', salaries)
    age_bins = np.arange(min(ages), max(ages) + 2, 1)
    salary_bins = np.arange(min(salaries), max(salaries) + 1, 1)
    t1.hist('Age', bins=age_bins, unit='year')
    plt.title('Age distribution')
    t1.hist('Salary', bins=salary_bins, unit='million dollars')
    plt.title('Salary distribution')

    histograms(full_data)
    print('Two histograms should be displayed below')
```

Two histograms should be displayed below

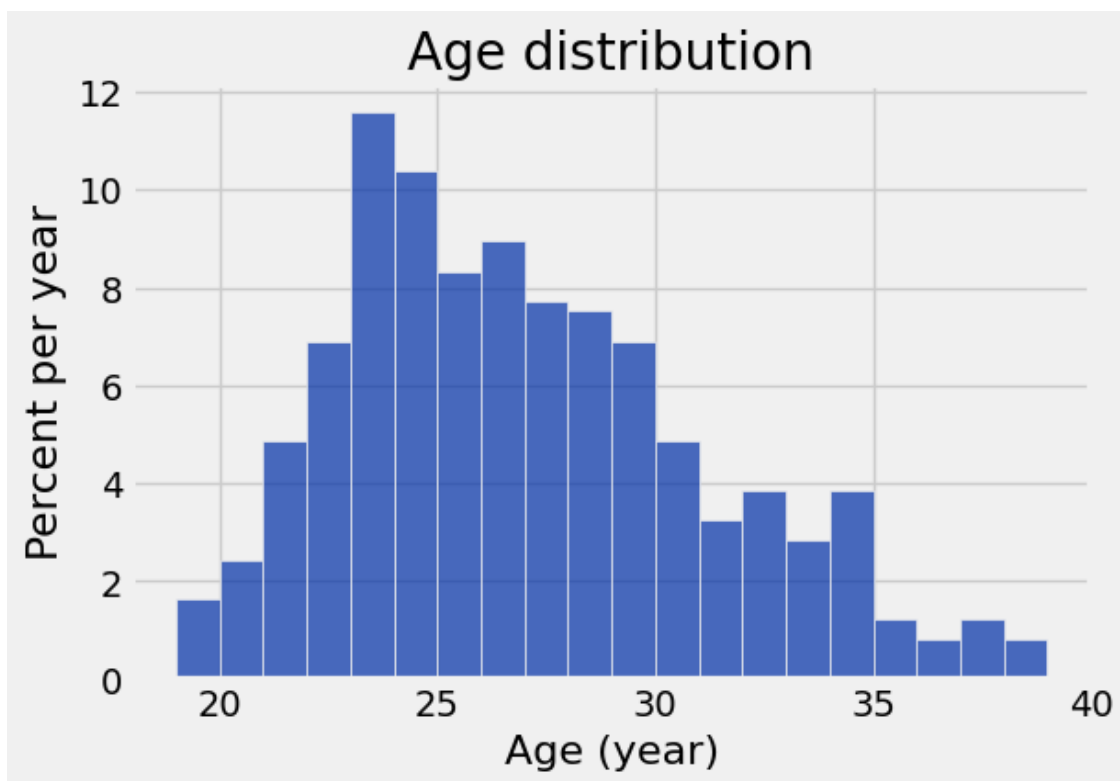


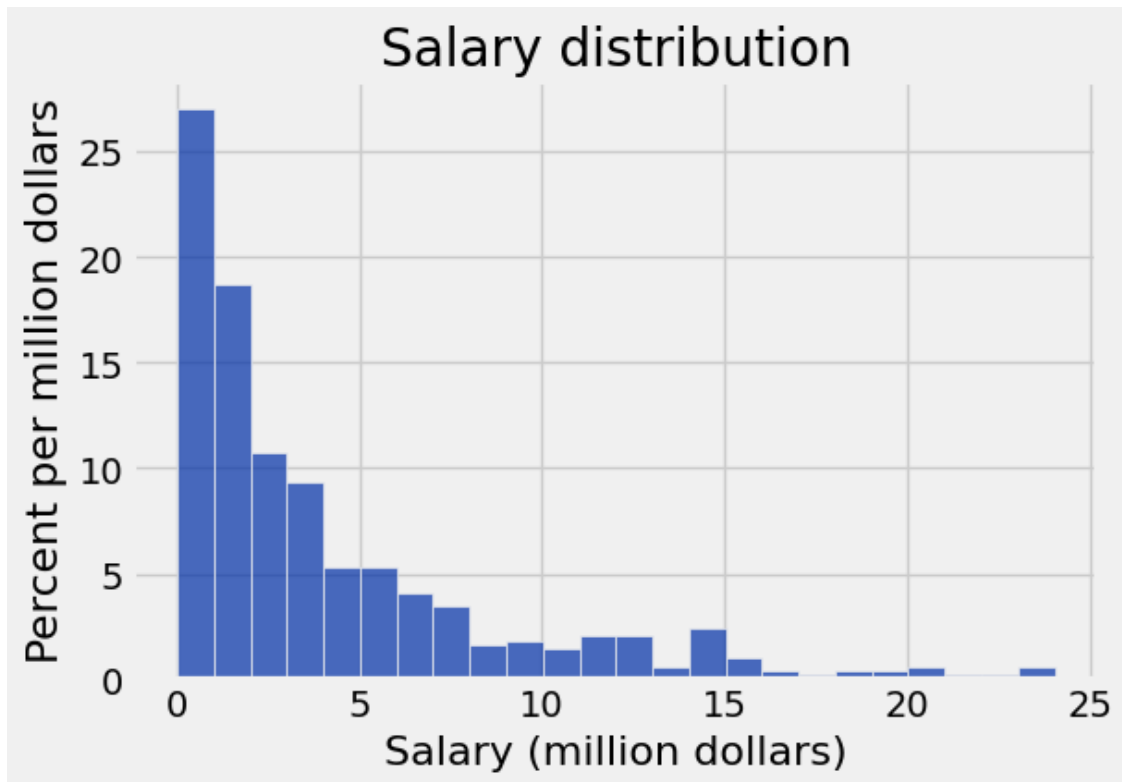
Question 3.1. Create a function called `compute_statistics` that takes a table containing an “Age” column and a “Salary” column and: - Draws a histogram of ages - Draws a histogram of salaries - Returns a two-element array containing the average age and average salary (in that order)

You can call the `histograms` function to draw the histograms!

```
[35]: def compute_statistics(age_and_salary_data):  
    histograms(age_and_salary_data)  
    age = np.mean(age_and_salary_data.column('Age'))  
    salary = np.mean(age_and_salary_data.column('Salary'))  
    return make_array(age, salary)  
  
full_stats = compute_statistics(full_data)  
full_stats
```

```
[35]: array([ 2.65365854e+01,  4.26977577e+06])
```





1.4 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab11(SSU).ipynb' --to pdf
```